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FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



Deep Learning and Its Applications (CO3133)

Semester 261 – Class CC01 – Group G-02

Assignment 2 – Milestone 1

Dataset Proposal

Tomato Leaf Disease Classification

Image Classification Track

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1 Dataset Name, Source, Version, and License

Dataset: Tomato Leaves Dataset, hosted by shreemunale/tomato-leaves-dataset on Hugging Face.

<https://huggingface.co/datasets/shreemunale/tomato-leaves-dataset>

Version: repository revision c029791b678863f6cb856589e4eea8d8f78e5931, accessed on 3 October 2026.

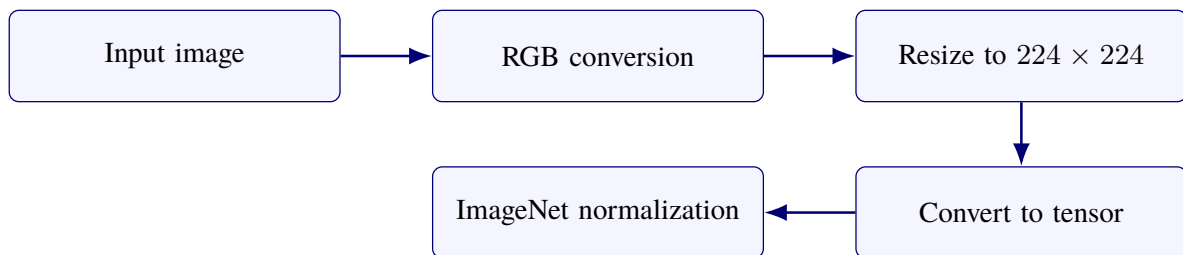
Origin and license: the dataset card describes a combination of tomato classes from PlantVillage and PlantDoc and lists CC BY 4.0.

2 Task, Input, and Output Definitions

Task: classify a tomato-leaf image into one of seven disease categories or the healthy category. The prediction unit is one image. The motivation is to recognize visual disease symptoms in leaf photographs.

Input: one RGB image, resized to 224×224 and represented as a tensor of shape $3 \times 224 \times 224$.

Output: eight class logits, with the predicted class obtained by taking the largest logit. Labels will follow the dataset card's zero-based order: bacterial spot (0), early blight (1), late blight (2), leaf mold (3), Septoria leaf spot (4), yellow leaf curl virus (5), mosaic virus (6), and healthy (7).



The same preprocessing will be used for all main configurations. Training-only augmentation will use mild cropping and horizontal flips; validation and test preprocessing will be deterministic.

3 Number of Samples and Annotation Types

The dataset card reports **17,996 images**: 14,404 training, 1,796 validation, and 1,796 test images across eight classes. Each image has **one image-level class label**.

These published counts meet the dataset size requirements: eight classes exceed the minimum of five, and 14,404 training samples exceed the minimum of 5,000. Each class has at least 59 test images in the published split.

4 Preliminary Distribution Analysis

Table 1 and Figure 1 are reproduced from the dataset card's numerical summaries.

Table 1: Published class counts from the Hugging Face dataset card.

Class	Train	Validation	Test
Bacterial Spot	1,930	235	241
Early Blight	966	131	117
Late Blight	1,710	210	209
Leaf Mold	999	120	125
Septoria Leaf Spot	1,771	210	223
Yellow Leaf Curl Virus	4,930	624	625
Mosaic Virus	509	66	59
Healthy	1,589	200	197
Total	14,404	1,796	1,796

Yellow leaf curl virus is the largest training class (4,930 images), while mosaic virus is the smallest (509 images), giving a ratio of $4930/509 \approx 9.69$. This imbalance motivates reporting macro-F1 alongside accuracy.

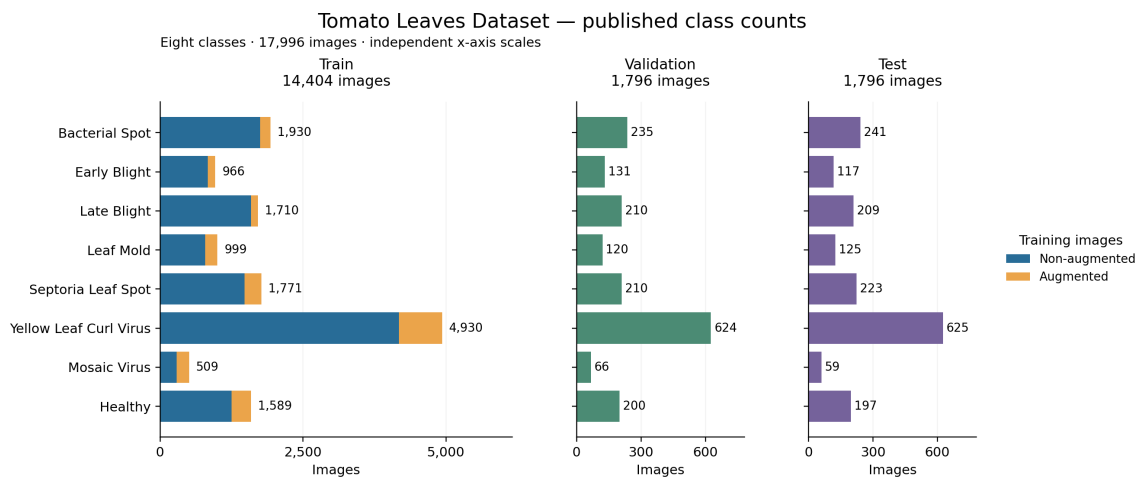


Figure 1: Published class counts. Training bars distinguish augmented and non-augmented images; the panels use independent horizontal scales.

5 Data-Splitting Plan

Start with the dataset’s provided train, validation, and test partitions. All models will use the same finalized split manifest. The training set will fit model parameters, the validation set will select checkpoints and settings, and the test set will provide the final evaluation.

If duplicate families cross the provided partitions, propose a corrected split with approximately 80% training, 10% validation, and 10% test images, using seed 42 and preserving class coverage where grouping allows. Record the revised counts and any deviation from these ratios. The correction will be documented before main experiments.

6 Split Unit Used to Prevent Leakage

Split unit: one original photograph and all its identifiable derivatives. Exact copies, augmented versions, and related leaf crops from the same source photograph must remain in one partition.

Use file hashes for exact duplicates and image similarity plus inspection for candidate near-duplicates. Use filenames or source metadata to recover original–derivative relationships where possible. Document unresolved source relationships.

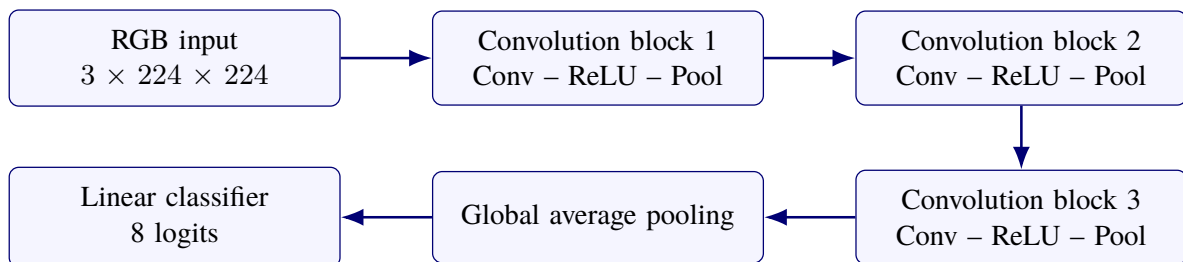
7 Evaluation Metrics

Required metrics: accuracy and macro-F1. Use validation macro-F1 to select the best checkpoint for every model. Report final test metrics under the same evaluation protocol.

Include per-class precision, recall and F1, a confusion matrix, and training/validation curves. Compare total and trainable parameters, training time, and inference time on the same hardware. Qualitative analysis will show correct and incorrect predictions and discuss common class confusions, difficult backgrounds, and minority-class failures.

8 Baseline Plan

Train a small CNN from scratch using three convolution–ReLU–pooling blocks, global average pooling, and an eight-logit linear classifier. Use cross-entropy loss on the logits.



The proposed initial training settings are Adam, learning rate 10^{-3} , batch size 32, twenty epochs, and seed 42. The baseline provides a simple reference for the benefit of pretrained features.

9 Planned Pretrained Model and Fine-Tuning

Use ImageNet-pretrained ResNet-18 and replace its classifier with Linear(512, 8). Compare the following two configurations:

Table 2: Controlled experiment: trainable backbone versus frozen backbone.

	Frozen backbone	Full fine-tuning
Backbone weights	Frozen	Trainable
Eight-class head	Trainable	Trainable
Initial weights	Same ImageNet weights	Same ImageNet weights
Head initialization	Same seed	Same seed

Hypothesis: updating pretrained features improves macro-F1, at the cost of additional training computation.

Changed factor: whether backbone parameters receive updates.

Fixed factors: data split, seed 42, preprocessing, augmentation, batch size 32, twenty epochs, Adam with learning rate 10^{-4} , constant learning-rate schedule, and validation checkpoint rule. Keep backbone BatchNorm running statistics fixed in both configurations to avoid introducing a second change through those statistics.

Decision criterion: compare validation macro-F1 at each configuration's selected checkpoint; a higher value is an observed improvement under this protocol. Report its percentage-point difference and training-time cost, then evaluate the finalized configurations on the test set.

10 Compute Estimate

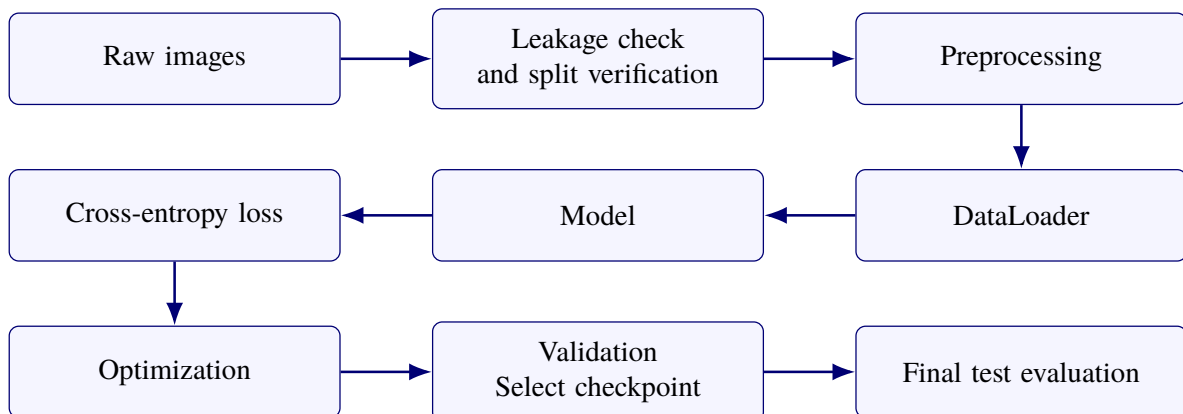
The planned experiments are designed to remain feasible on a single NVIDIA T4 GPU through Google Colab. The small CNN baseline, frozen-backbone ResNet-18, and fully fine-tuned ResNet-18 will be trained sequentially.

With 224×224 inputs, batch size 32, and 20 epochs per configuration, the initial compute budget covers three training runs, totaling 60 epochs. Training and inference times will be recorded for each configuration.

11 Subset Selection Rules

Use all eight classes and all valid images in the chosen repository revision, subject to the documented leakage and integrity checks. Record any exclusions and their reasons. If a reduced dataset becomes necessary, document explicit selection rules, class counts, and a seed before using it for main experiments.

12 Proposed Workflow





13 References

1. Tomato Leaves Dataset, Hugging Face, repository revision c029791, accessed 3 October 2026. <https://huggingface.co/datasets/shreemunale/tomato-leaves-dataset>
2. D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, "PlantDoc: A Dataset for Visual Plant Disease Detection," CoDS-COMAD, 2020. <https://arxiv.org/abs/1911.10317>
3. K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, 2016.